

Encounters with Visual Misinformation and Labels Across Platforms

An Interview and Diary Study to Inform Ecosystem Approaches to Misinformation Interventions

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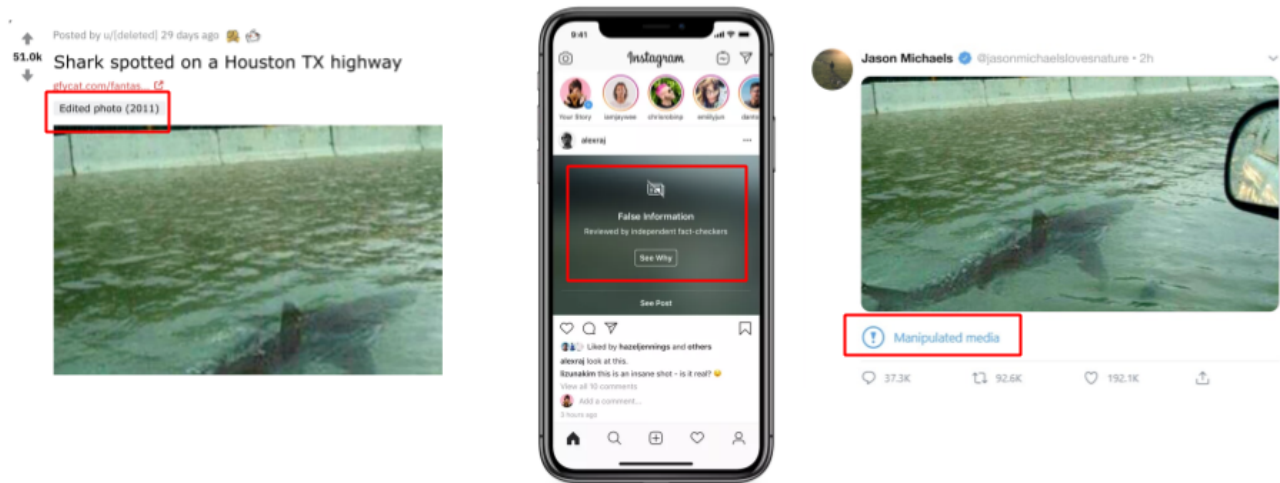


Figure 1: Label stimuli applied to a doctored photo of a shark on a highway [8] from left to right: user-applied labels (Reddit), fact-checker overlay labels (Instagram), “Manipulated Media” label without overlay (Twitter).

ABSTRACT

Social media platforms face rampant misinformation spread through multimedia posts shared in highly-personalized contexts [10, 11]. Foundational qualitative research is necessary to ensure platforms’ misinformation interventions are aligned with users’ needs and understanding of information in their own contexts, across platforms. In two studies, we combined in-depth interviews (n=15) with diary and co-design methods (n=23) to investigate how a mix of Americans exposed to misinformation during COVID-19 understand their information environments, including encounters with interventions such as Facebook fact-checking labels. Analysis reveals a deep division in user attitudes about platform labeling interventions, perceived by 7/15 interview participants as biased and punitive. As a result, we argue for the need to better research the unintended consequences of labeling interventions on factual beliefs

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CHI ’21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

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ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451807>

and attitudes. Alongside these findings, we discuss our methods as a model for continued independent qualitative research on cross-platform user experiences of misinformation in order to inform interventions.

CCS CONCEPTS

• **General and reference;** • **Human-centered computing** → **User centered design;** *Participatory design;* **User studies;** • **Information systems** → **Social networks;**

KEYWORDS

visual misinformation, manipulated media, misinformation, news media, social media, platform design

ACM Reference Format:

Emily Saltz, Claire Leibowicz, and Claire Wardle. 2021. Encounters with Visual Misinformation and Labels Across Platforms: An Interview and Diary Study to Inform Ecosystem Approaches to Misinformation Interventions. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI ’21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3451807>

Whether conspiratorial videos on YouTube, political memes on Instagram, or viral protest photos on Twitter, images and videos on technology platforms have immense power to influence public

opinion [7, 14]. As technology platforms moderate the credibility of multimedia content, they are not only tasked with evaluating whether or not content is manipulated and/or misleading, but also with determining how to respond to their evaluations.

Alongside interventions like content removal or downranking content, platforms employ labels on posts as a strategy to address misinformation. We define “label” here as any textual or visual cue added to a post with the goal of informing users about the credibility of its content. Labels today take many forms, from contextual information such as Twitter’s labeling of state [19] and manipulated media [21] to Facebook’s evolving set of labels based on fact-checker ratings of the veracity of claims in a post [9]. While such interventions have an intuitive appeal as “more nuanced” [13] approaches for fortifying media integrity while preserving freedom of expression, compared to the blunt “leave up/take down paradigm” approach for post removal [4], it is crucial for researchers to be able to independently evaluate the effects of labeling, and other misinformation interventions, in practice. Platforms are also best positioned to provide ecologically valid experimental data about the effects of interventions, yet currently fail to provide sufficient information to enable independent evaluation by other researchers [16]. In this paper, we aim to explore new ways for researchers to empirically understand users’ perceptions and attitudes towards evolving labeling interventions on multimedia misinformation.

1 LIMITATIONS OF EXISTING INTERVENTION RESEARCH

As Morrow and colleagues’ (2021) review of content labeling describes, there is a robust social science literature on the effects of labels, yet it is unclear to what extent many of these findings generalize to real world contexts. For example, one frequently-cited finding from this literature is “the implied truth effect” from Pennycook and colleagues (2020) which posits that “attaching warnings to a subset of fake news headlines increases perceived accuracy of headlines without warnings” [17]. If this finding generalizes, it poses a profound existential threat for the premise of labeling content based on veracity as an effective means to reduce belief in misinformation.

Additionally, while behavioral research often aims to abstract away from the complexities of platform environments to underlying psychological principles, changing context-specific elements, such as social metrics, has been found to have effects on behaviors related to misinformation, as in the finding from Avram and colleagues (2020) that, “people are more vulnerable to low-credibility content that shows high levels of social engagement” [1]. That means that many existing experimental findings are subject to change if encountered in naturalistic contexts, where contextual factors, such as social engagement in a user’s specific networks, become relevant to the effect of any given intervention.

2 TOWARD ECOLOGICALLY VALID INTERVENTION TESTING

In contrast, while platforms appear to conduct ecologically valid research to inform their intervention decisions, they may be hesitant to share results of this research beyond high-level blogs in coordination with public relations teams [18]. These efforts fail to meet the

needs of researchers attempting to independently evaluate misinformation, as described in a paper by Pasquetto and 42 other authors (2020) outlining how they would benefit from additional access to social media data [16]. The lack of sufficient information sharing stifles the evaluation of misinformation interventions, leading to a breakdown where platforms are understandably wary to uncritically adopt recommendations from studies that are not ecologically valid for their users. In turn, independent researchers are unable to design the most relevant, ecologically valid experiments about interventions when lacking the most current information about what platforms have already studied, as well as realistic environments in which to conduct testing.

Given the present limitations, we explore a methodology as an attempt to balance the above challenges in studying platform interventions. We combine findings from an interview study with controlled stimuli alongside a naturalistic diary study to shed light on how multimedia misinformation and interventions are perceived.

3 METHODS

To better understand how Americans react to visual information and misinformation interventions, we conducted two qualitative studies focused on U.S. news topics: an interview study with 15 participants about visual misinformation and existing labels (S1), followed by an 11-day diary study with 23 additional participants in which we analyzed 121 news-related posts (average of 5 posts per person) that elicited strong reactions across platforms (S2). Both studies were recruited and conducted in English remotely through dscout [5], an online qualitative research platform. All participants signed terms of service and consented to each study, with additional consent language written in consultation with dscout project managers and legal counsel. Semi-structured interview protocols and diary instructions and prompts are provided in supplementary materials.

3.0.1 Study Participants. We recruited a broad mix of Americans who we believed regularly saw low credibility information on digital platforms, in order to ensure we were talking to people more likely to encounter the types of visual misinformation we were studying. Given the early discovery phase of this research, we prioritized depth of analysis per interview over quantity of interviews in order to better define our target populations for future study.

In S1, we recruited 15 American English-speaking participants. To screen for participants we believed were regularly exposed to digital misinformation, we relied on self-reported news consumption of at least one low credibility site on the Iffy Index of Unreliable Sources [15], an index which includes only sites with low factual-reporting levels as determined by the Media Bias/Fact Check methodology [12]. 15/15 participants self-reported media consumption from at least one Iffy Index source. Additionally, all participants self-reported seeing and sharing information about COVID-19 on social media at least weekly. All had accounts with at least one social media platform (Facebook, Instagram, Twitter, or Reddit) relevant to examples in the study, and reported using these platforms daily or more. Participants included a mix of genders (7 female, 8 male), race/ethnicities (3 Asian, 1 Black, 2 Hispanic, 1 Middle Eastern or North African, 8 White), ages (20–63), locations (6 urban, 9 suburban), income levels (Less than \$25,000–Over \$150,000),

education levels (high school graduate to post-graduate college), and politics, skewing Republican (8 Republicans, 4 Democrats, 3 Independents).

S2 featured 23 participants chosen with similar recruiting criteria. Based on findings from S1 that many people had not personally encountered labels, we also added a screening question about past exposure to labels to ensure that all participants had reported seeing labels at least “sometimes (a handful of times but not regularly)” or more. Additionally, based on observing strong partisan trends in attitudes in S1, we more explicitly sought partisan balance to better understand these differentiators (8 Republicans, 7 Independent/Other, 8 Democrats). Participants were a mix of genders (11 female, 12 male), race/ethnicities (4 Asian, 6 Black, 4 Hispanic, 1 Middle Eastern or North African, 8 White), ages (20–63), locations (9 urban, 13 suburban, 1 rural), income (Less than \$25,000–Over \$150,000), and education levels (high school graduate to post-graduate college). 23/23 self-reported media consumption from one or more Iffy Index source. See supplementary materials for more participant details.

3.0.2 Interview Study Procedure (S1). S1 aimed to unearth how people understand existing terminology for visual misinformation and labeling interventions. The 60-minute interviews were conducted in four parts. We began with a semi-structured interview about users’ media habits. Next, interviews featured a think-aloud discussion and comparison of images of existing manipulated media interventions from Instagram, Reddit, and Twitter shown in sequence in a Google slideshow, all as applied to an edited image of the highway shark shown in Figure 1. We continued with a series of label terminology and photo/video elicitation tasks for a random selection of five fact-checked image and video examples corresponding to distinct manipulation types as defined by the MediaReview schema [2], as shown in Figure 2. These examples were chosen because they represented actual instances of fact-checked posts on common topics (COVID-19 or the U.S. election) that we expected to be relevant to all of our participants. We randomized examples in Google Slides using the Slides Randomizer add-on in order to control for ordering effects. This exercise involved think-aloud discussion about each post and the participant’s ideal terminology and intervention. Examples were shown alongside lightly-edited, original fact-checker descriptions, alongside a set of terminology options relevant to each rating type. These tasks were not intended to provide a perfectly realistic simulation of a platform environment, but rather to enable us to elicit attitudes on a controlled yet realistic variety of examples in order to later compare with more personalized reporting of actual behaviors and content encountered in S2. Finally, interviews ended with open-ended questions to understand how participants feel about platforms labeling media, and why.

To analyze interview data, a moderator and a notetaker debriefed notes by interview section and summarized takeaways after each interview. We then worked through these notes and transcriptions of approximately 15 hours of audio to synthesize findings through bottom-up thematic analysis [3] to iteratively generate codes for various interview notes and utterances, grouping these codes into successively higher-level themes concerning attitudes toward visual information and labeling interventions. Key themes are presented alongside findings from S2 in Findings and Discussion below.

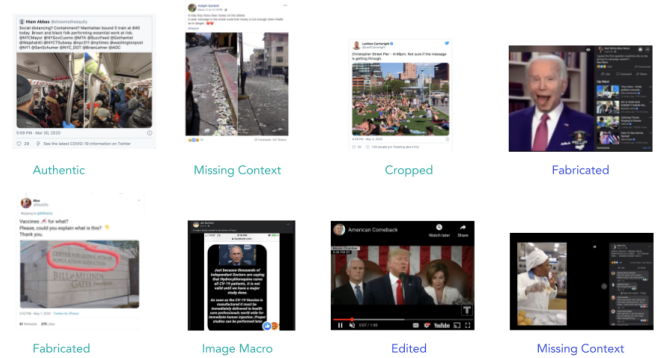


Figure 2: Images (green) and videos (blue) shown to participants in A3 of S1 with categories based on MediaReview[2] fact-checker ratings.

3.0.3 Diary Study Procedure (S2). The diary study aimed to build upon and complement findings from S1 by capturing critical moments driving trust and distrust in images and videos in context of their everyday use across platforms, and users’ mental models of how the platforms they use moderate visual information and apply interventions, in the context of their own feeds.

To capture in-context experiences, we used dscout’s mobile diary tool. The diary began with one entry of background questions to better socially situate participants and understand their news habits, such as “how [they] curate [their] social media feeds for [their] go-to platforms” and why. In the next section, spanning seven days, participants were asked to log at least five screenshots and selfie-style videos describing personal encounters with a range of visual information in real time, across platforms. We used two triggers for logging images or videos from their personal contexts: 1) “an image or video post about any current news event [that made them] pause from a strong reaction of any kind” or 2), any time they saw a label applied by a platform like Facebook, Instagram, or Twitter, such as “false information” or “manipulated media” labels. We used “strong reaction” as a trigger to ensure we captured a range of posts that users strongly supported, questioned, or opposed, in order to gain a holistic sense of the meaningful influences in their lives for understanding current events—not assuming that platform classifications of credibility would align with their own. Finally, the diary ended with participants reflecting on their entries and selecting an issue to focus on in order to sketch an alternative user experience for their platform(s) of choice. Preliminary observations of media entries and sketches are presented to complement and build on key themes from S1 in Findings and Discussion.

4 FINDINGS AND DISCUSSION

Preliminary findings from both studies show deeply divided attitudes toward platform interventions. Some experience platform interventions as patronizing and political, targeting their preferred media and autonomy to understand content for themselves; others see interventions like labels as positive, if insufficient steps, to clarify media credibility, especially for those they perceived as more gullible others.

4.1 Visual misinformation posts as part of larger divisive narratives

In our initial interviews, many participants described feeling exhausted by an overwhelming deluge of conflicting, negative, emotionally-charged content on social media (S1-P06, S1-P11, S1-P15). Reactions to misleading and false image and video posts were just manifestations of broader feelings of discontent about the polarized state of news media and current events—in other words, media format was a less salient factor than the narrative and sociocultural framing of the post (e.g., pro-mask vs. anti-mask). It was from this polarized news environment that labels on posts are seen by many as further aggravating partisan divisions.

4.2 Label Attitude A: Platforms are responsible for fact-checking

Many believe it's the platform's responsibility to be clear and objective about what's true or false (8/15 in Study 1; four Democrats, three Republicans, one Independent), saying: "If it's clearly fake, mark it fake for others" (S1-P05); "Platforms need to take social responsibility" (S1-P14). Of the 14 label treatments logged in our diary study, six labels were viewed as appropriately applied (two labels each from Democrats S2-P01 and S2-P14; two labels total from Republicans S2-P22, S2-P08). Several diary entries of labels expressed this sentiment, including one Republican participant who encountered a labeled Tucker Carlson video on Facebook:

"It was really surprising to see something like a huge personality like [Tucker Carlson] would release something that is really not true...I really felt that I had to read more about it. I am glad Facebook marked it as false." –S2-P22

4.2.1 Stronger labels for stronger harms. Many pro-label participants desired stronger labels for stronger perceived harms, such as those envisioned from a photo of the Bill and Melinda Gates Center (as shown in Figure 2) which was edited to read "Center for Global Human Population Reduction:" "Once something is out there, it's hard to unsee it" (S1-P07); "If you just have a label, can be easy to miss" (S1-P15); "if it's not obscured and they don't take the time to read the verification, the damage is already done" (S1-P14).

4.2.2 If it's 'fake,' remove it. In severe cases, removal was sometimes seen as preferable to labeling. Of 10 entries rated by nine participants as "completely inaccurate," people wanted the media removed in a majority of instances (6/10). For example, in response to a conservative Instagram meme with a false claim that George Floyd died of a drug overdose:

"If Instagram knows that these claims are just completely false, it needs to be taken off the app. Why would you allow something that is incorrect to keep spreading? Just because you put a warning doesn't mean people who already subscribe to that ideology aren't going to believe it." –S2-P01



Figure 3: Screenshot of a fact-checking label applied to a video posted on Facebook by Tucker Carlson, as encountered by S2-P22.

4.3 Label Attitude B: Platforms are policing free speech with bias

In contrast, many viewed labeling (7/15 in S1; four Republican, three Independent) as judgmental, paternalistic, and against the platform ethos. Some made comparisons to platforms as police: "I don't want Instagram and Twitter to be the police. It's on me to understand the facts within these posts...would they still be doing this if someone else was in the White House?" (S1-P08). This was seen by some as at odds with the intent of social media: "It's an open forum, therefore it's its nature" (S1-P12). This hostile attitude toward labeling was also seen in the diary study: 7/14 labels submitted were understood as inappropriately applied. All of the participants submitting these examples were Republican (one label each from S2-P02, S2-P05, S2-P16, four labels from S2-P03). For example, one participant encountered a post that he independently investigated and determined was accurate as originally posted:

"When I see fact-checkers on a post, it agitates me. Because they think [they] know it all. The original post was correct. I've seen a lot of fact-checking done [based] on personal opinion, using fake articles to support their agenda." –S2-P16

4.3.1 False positives: Label severity mismatched with harm. Others also noted times they felt labels were inappropriately applied without taking into account the poster's intent. One participant, S1-P04, described two times she felt she had personally had labels applied to her posts without merit: The first post was a meme that she felt "was very obviously a joke." In response to the label, she said she was "livid." The second time she knowingly shared a false post, but it was in order to educate others: "I had actually in my post shared this picture and in my own words said 'this is why you need to fact-check.' And they still blocked it!...It's like they look at the pictures or the headline that you're posting and not look at what you post with it." Both experiences reduced her trust in the accuracy of fact-checking labels applied by Facebook.

4.3.2 Overlays as censorship. Some felt the overlay style could be particularly punitive and patronizing, described by several participants as “offensive” or “censorship,” saying: “Instagram is ‘the most offensive’...people are smart enough to understand whether [the image in Figure 1 is] a shark or not (S1-P12); “Why do I need somebody to tell me that something’s false? Why not just let me find out if it’s false or not. I don’t want somebody to tell me” (S1-P09); “I want to decide for myself whether the info being blocked is true or false through my own lens” (S1-P11).

4.3.3 The mythical “unbiased” label. Many expressed the desire for a “common ground” (S1-P15) neutral authority or third-party that all would trust, rather than platforms seen as profit-driven and politically-biased, picking and choosing what to check. For some, fact-checkers themselves were seen as biased: “It’s really a moral fact-checker, it’s not really facts...What’s their agenda? Who’s paying them?” (S1-P09). What is “neutral?” Several wished for automated interventions that could be ostensibly applied without personal bias, across all content. Some suggested, “Have the algorithm take care of it” (S1-P06). Others felt neutrality could be accomplished if a story was corroborated by news outlets with diverse perspectives: “If it’s saying the same on the left and right I know I can trust it” (S1-P12).

4.4 Design Opportunities

Beyond current labels, what other forms might interventions take? S2 participants sketched concepts in response to the visual misinformation issues and platforms they felt were most important in their own lives. Most built on strategies currently employed by social media platforms, such as: better fact-checks, including additional source information and ability to check at will, across platforms (9 responses) and better filters, including filtering only for confirmed “verified” news media (6 responses). Additional strategies included: centering user perspectives through crowdsourcing (3 responses), editing feeds to include less negative, divisive media overall (2 responses), or improving platform appeal processes (2 responses), and external oversight processes (1 response). Notably, top strategies located the problem as a failure of platforms to accurately categorize information as “true” and/or “false” in a way that was aligned with their own definitions.

5 CONCLUSION AND FUTURE WORK

These findings have profound implications for the study of platform interventions. Though drawing from a limited sample size, both studies vividly demonstrate the intense emotional reactions evoked by labels for misinformation in general—perceived by many as overly paternalistic, biased, and punitive. As such, current post-level labeling interventions seem likely to continue to provoke feelings of distrust and hostility toward platforms and content correction labels. Beyond this, the question remains for future research: how do these attitudes relate to likelihood to update beliefs based on a correction label?

Some shortcomings of our methodology included a lack of a clear way to map between findings for each manipulated media example in S1 to the types of examples submitted in S2. In the future, we would recommend working with the same user group in S1 and S2, and perhaps starting with the diary study first in order to enable

better reflection and probing on both the specific entries submitted and the way they contrasted to the additional sets of examples. Additionally, though we wanted to keep our prompt open in order to be able to understand how labeled media compared to other meaningful news media influences, the breadth of entries made it hard to analyze attitudes meaningfully. In a future study, it could be valuable to focus exclusively on labels submitted over a longer period of time to better detail these reactions and identify more specific pain points and experiences with a range of interventions, including changes in beliefs over time.

As anticipated, our qualitative methodology complicated discussions around established experimental findings about interventions, in particular “the backfire effect”—the idea that “when a claim aligns with someone’s ideological beliefs, telling them that it’s wrong will actually make them believe it even more strongly” [20]. Though this phenomenon is thought to be rare [20], our findings suggest that emotionally-charged, defensive backfire reactions may be common in practice for American social media users encountering corrections on social media posts about news topics. While our sample size was too small to definitively measure whether the labels actually strengthened beliefs in inaccurate claims, at the very least, reactions described above showed doubt and distrust toward the credibility of labels themselves—often with reason, as in the case of “false positive” automated application of labels in inappropriate contexts. As for other past findings, such as “the implied truth effect,” [17] it’s notable most participants felt the quality of their social media feeds was generally mixed, and understood that they were not thoroughly checked for accuracy. This would indicate that “the implied truth effect” is likely not a severe concern for the current types of labels in use, though this too requires a more thorough analysis.

At a minimum, these insights should push platforms, researchers, and journalists to more holistically understand the feelings of discontent and uncertainty that lead many to engage with misinformation in alternative media narratives in the first place, and invest in ways to legitimately earn the trust of consumers disillusioned with these institutions. Specific design approaches could include topic-level “pre-bunking” [6] to clarify larger misleading narratives. Our findings predict that this type of intervention, which does not explicitly target a particular user or post and provides exploratory information, would be valuable for both Label Attitude A, who often thinks they don’t need content labels for themselves, as well as Label Attitude B, who want to be informed but are suspicious that content-labels are applied unevenly with bias. In addition, since those with Label Attitude B resented the perceived paternalism of labels, they may be more receptive to corrections through more community-centered moderation, or collaborative information hubs in the model of Wikipedia.

In summary, our findings point to major limitations in current visual misinformation labeling approaches as experienced by end-users. As a next step, a survey could serve as a supplement to initial qualitative studies, assessing emerging themes from interviews and diary submissions to better understand how factors like political affiliation, news habits, social media behaviors, and demographics correlate with intervention attitudes. In addition, there are fruitful opportunities for any researcher to use qualitative methods to work

closely with the most affected platform users to define context-specific needs. Doing so will bolster the efficacy of emotionally resonant interventions for visual misinformation and information at large.

ACKNOWLEDGMENTS

Thanks to Victoria Kwan, Tommy Shane, Laura Garcia, and Pedro Noel at First Draft, and Steven Adler, Nicholas Anway, Kemi Bello, Jeffrey Brown, B Cavello, Riccardo Fogliato, Adam Schetky, Jonathan Stray, Tina Park and all of our colleagues at the Partnership on AI, the AI & Media Integrity Steering Committee, and The Duke Reporters' Lab.

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